

The Implication of Brown Adipose Tissue for Humans

We here discuss the role of brown adipose tissue on energy homeostasis and assess its potential as a target for body weight management. Because of their high number of mitochondria and the presence of uncoupling protein 1, brown fat adipocytes can be termed as energy inefficient for adenosine-5'-triphosphate (ATP) production but energy efficient for heat production. Thus, the energy inefficiency of ATP production, despite high energy substrate oxidation, allows brown adipose tissue to generate heat for body temperature regulation. Whether such thermogenic property also plays a role in body weight regulation is still debated. The recent (re)discovery of brown adipose tissue in human adults and a better understanding of brown adipose tissue development have encouraged the quest for new alternatives to treat obesity since obese individuals seem to have less brown adipose tissue mass/activity than do their lean counterparts. In this review, we discuss the physiological relevance of brown adipose tissue on thermogenesis and its potential usefulness on body weight control in humans.